

Common Cyber Attacks in Cyberpreneurship

- ▶ Ransomware Attack
- ▶ Distributed Denial of Service (DDoS)
- ▶ SQL Injection
- ▶ Business Email Compromise (BEC)
- ▶ Account Hijacking
- ▶ Identity Theft
- ▶ Carding Attack
- ▶ Website Defacement
- ▶ Insider Attack
- ▶ Social Engineering

Ransomware Attack

- ▶ Malware locking business files and requiring payment to unlock them.

Distributed Denial of Service (DDoS)

- ▶ Overloading website with fake traffic making real customers unable to access.

SQL Injection

- ▶ Injecting malicious commands into website fields to steal or alter database data.

Business Email Compromise (BEC)

- ▶ Attacker impersonates business owner through email to request money or sensitive data.

Account Hijacking

- ▶ Hackers seize control of business social media accounts used for sales.

Identity Theft

- ▶ Using brand name or logo to create fake pages that scam customers.

Carding Attack

- ▶ Using stolen card details to make fraudulent purchases on an ecommerce site.

Website Defacement

- ▶ Changing website appearance to embarrass or scare customers.

Insider Attack

- ▶ Trusted staff or contractor misusing access to steal or leak data.

Social Engineering

- ▶ Social engineering is the act of tricking people into revealing confidential information. In cyberpreneurship it harms online businesses by exploiting human trust to steal access or money. Examples are:
 - ▶ Phishing
 - ▶ Spear Phishing
 - ▶ Whaling
 - ▶ Vishing
 - ▶ Smishing
 - ▶ Pretexting
 - ▶ Baiting
 - ▶ Tailgating

Phishing

This is a social engineering attack where cybercriminals send fake messages designed to look like legitimate communication in order to trick victims into revealing confidential information.

- ▶ **Spear Phishing:** targeted phishing attack aimed at a specific person or business.
- ▶ **Whaling:** phishing attack aimed at high-ranking executives or business owners.
- ▶ **Vishing:** voice call phishing where attackers deceive victims over the phone to get information.
- ▶ **Smishing:** phishing delivered through SMS or messaging apps to steal login or payment details.

Pretexting

This is a social engineering attack where the attacker creates a false identity or scenario to convince the victim to reveal confidential information.

Baiting

This is a social engineering attack where the attacker uses a tempting offer such as a free file or reward to lure the victim into downloading malware or giving away sensitive data.

Tailgating

This is a social engineering attack where an unauthorized person enters a restricted area by closely following an authorized person without proper identification or access rights. It takes advantage of lapses in access control and physical security checks.

Cybersecurity

This is the practice of protecting computer systems, networks, digital devices, data, and online services from unauthorized access, attacks, damage, or theft. It involves the tools, policies, technologies, and processes used to prevent, detect, and respond to cyber threats so that information and digital resources remain safe, confidential, accurate, and available for use.

Cybersecurity Practices

► Use Strong Passwords and MFA

A cyberpreneur must use strong, unique passwords for all accounts (email, business dashboard, social media, payment gateway). In addition, Multi-Factor Authentication (MFA) should be enabled so that even if a password is stolen, the attacker cannot log in without the second factor like OTP or authenticator code.

Cybersecurity Practices

► Update Systems, Software and Plugins Regularly

Websites, apps, browsers, CMS plugins and operating systems must always be kept up-to-date. Updates contain security patches that fix weaknesses hackers can exploit. Running outdated software exposes the business to attacks such as ransomware, website defacement and data theft.

Cybersecurity Practices

► Backup Business Data Frequently

The cyberpreneur must create regular backups of important files, customer records, and business databases. These backups should be stored offline or in secure cloud storage. Backups allow the business to recover quickly if an attack occurs, especially in the case of ransomware.

Cybersecurity Practices

► Use HTTPS and Encryption

The website must run on HTTPS to protect customer data as it travels between the user and the server. Sensitive information such as customer records, payment details and business secrets should be encrypted so that even if attackers access the data, they cannot read it.

Cybersecurity Practices

► Use Secure and Trusted Payment Gateways

Cyberpreneurs should only use verified payment gateways like Paystack, Flutterwave, Stripe or PayPal to reduce risks of credit-card fraud and carding attacks. Secure gateways have fraud-detection tools and reduce the burden of storing raw card details.

Cybersecurity Practices

► Install Anti-Malware / Endpoint Protection

Anti-virus or endpoint security solutions should be used to detect and block malware such as ransomware, spyware and trojans. This helps prevent unauthorized software from running on the business devices.

Cybersecurity Practices

► Use a Web Application Firewall (WAF)

A Web Application Firewall helps protect the website from attacks like SQL injection and DDoS. It filters malicious traffic before it reaches the site. This is especially important for e-commerce or platforms that take customer inputs.

Cybersecurity Practices

- ▶ **Train Yourself and Staff on Phishing and Social Engineering**

People are often the weakest link. Cyberpreneurs must educate their workers and themselves to avoid clicking suspicious links, messages, or attachments. Training helps stop hackers who try to deceive staff to gain access to accounts or data.